Noise Reduction of a Mixer Grinder using Active Noise Cancellation

Project Abstract

A) Team & Institutional Context

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B) Problem & Idea Framing

Problem

One common problem with a mixer grinder is that it makes a lot of noise while in use. This mainly happens because the motor runs at a very high speed. The blades also make noise when grinding ingredients inside the jar. Sometimes the mixer shakes while running, which creates extra sound, and loose parts or old bearings can make the noise even worse. This loud sound can disturb people at home, especially when the mixer is used early in the morning or late at night. Reducing the noise of a mixer grinder can therefore make it more comfortable and convenient to use.

Proposed Solution

The idea is to reduce the noise produced by a mixer grinder using sound waves. When the mixer grinder is switched on, a sensor or microphone detects the sound produced by the motor and blades. The system then analyzes the sound waves and identifies their frequency. Based on this, a circuit generates an opposite sound wave that helps cancel the original sound. When both waves meet, they reduce the noise through a process called **active noise cancellation**. The aim is to make the mixer grinder quieter without affecting its normal working. However, the system needs to be designed carefully because the noise changes depending on the speed and the ingredients being ground.

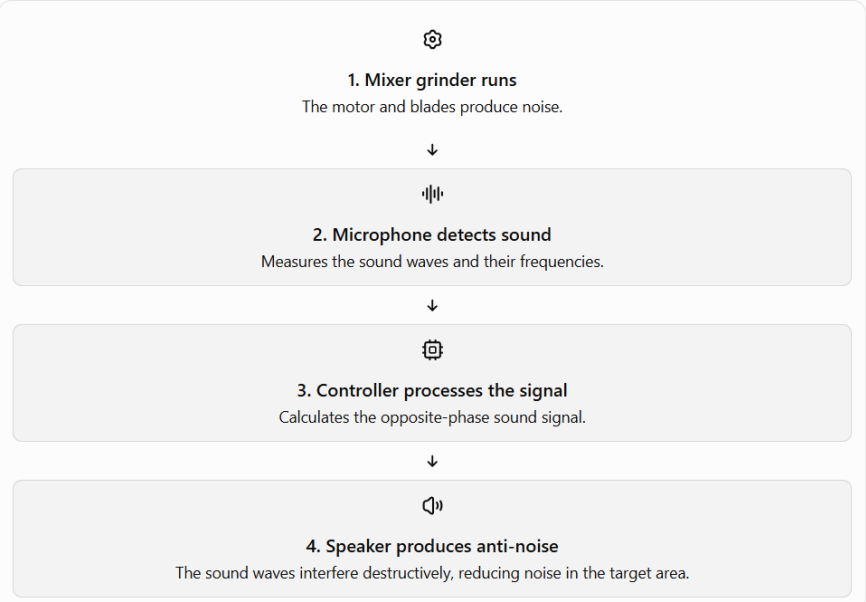


Figure 1: Working flow of the proposed noise-cancellation system

Science Behind It

- **Original wave:** rises and falls.
- **Opposite (antiphase) wave:** falls and rises at the same time.

- **Result:** the waves cancel each other partially or completely under suitable conditions.

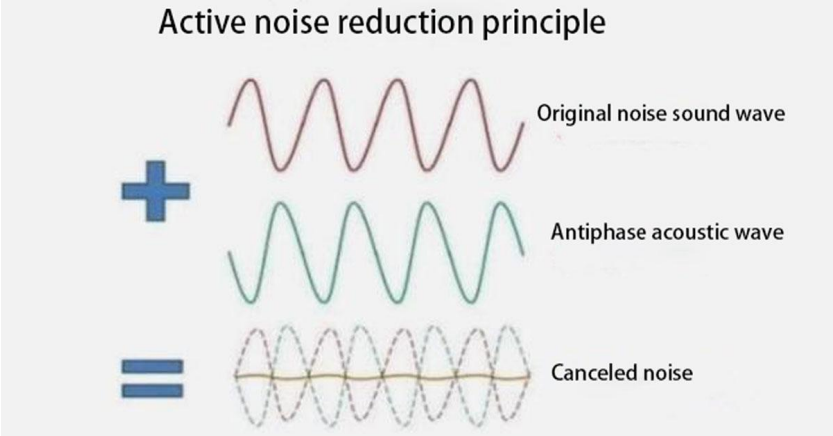


Figure 2: Active noise reduction principle (destructive interference)

C) Technical Feasibility

Challenge in the idea

A sensor can detect sound, but it cannot automatically identify the exact physical component producing it just by measuring the sound wave. Identifying the source may require multiple microphones, vibration sensors, or signal analysis.

Additionally, anti-noise works best at specific locations and frequencies. A mixer grinder produces complex, changing noise, so cancelling all of it throughout a room is difficult. A practical first prototype could target the **dominant low-frequency motor noise near the user's listening position**.

D) Execution Feasibility

Components

Microphone sensor module, controller / processor, audio amplifier, speaker, and a sound-level meter (for measuring the reduction achieved).

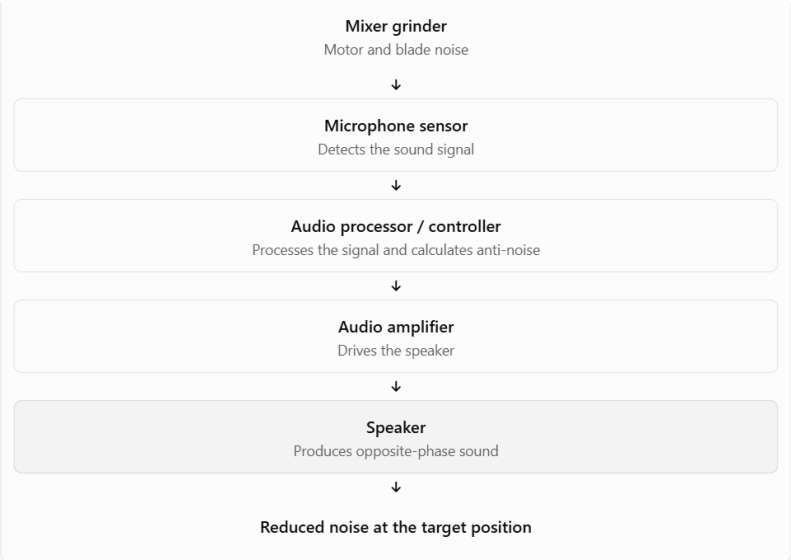


Figure 3: Block diagram of the prototype hardware

Estimated Cost

Rs. 3,000 – Rs. 8,500 (approx. INR)

Risk Assessment

Component / Task	Risk Level
ESP32 controller	Low
Microphone modules	Low
Speaker and amplifier	Low
Audio interface / DSP	Medium
Custom ANC circuit	High